

Aircraft Design Optimization Report

1. Problem Understanding

The task is to optimize a simplified aircraft design using an evolutionary algorithm. Each design is described by six variables: wing span (b), wing area (S_w), sweep angle, thickness-to-chord ratio (t_c), fuselage length (L_f), and fuselage diameter (D_f).

The three objectives — minimize drag, maximize fuel volume, maximize cargo volume — conflict with each other. A larger wing stores more fuel but can increase drag. A wider fuselage increases cargo space but also raises drag. Because no single design can be best at everything, the algorithm returns a Pareto front of trade-off solutions instead of one answer.

2. Algorithm Design

I used a genetic algorithm with Pareto-based selection and crowding distance for diversity.

Each solution is a real-valued vector $x = [b, S_w, \text{sweep}, t_c, L_f, D_f]$. The population is initialized by sampling each variable uniformly within its bounds. Solutions are evaluated using the notebook equations for C_D , V_{fuel} , V_{cargo} , and AR . All objectives are written as minimization: $[C_D, -V_{\text{fuel}}, -V_{\text{cargo}}]$.

Constraints are handled with a squared penalty function — larger violations are penalized more. Feasible solutions always beat infeasible ones in any comparison, regardless of objective values.

Selection is binary tournament. Crossover is arithmetic blending of two parents. Mutation adds Gaussian noise to each variable independently, clipped to bounds.

Survivor selection follows the NSGA-II approach: parents and children are merged, sorted by Pareto rank, and filled rank by rank into the next generation. When a rank doesn't fully fit, crowding distance selects the most spread-out solutions. This is how diversity is maintained.

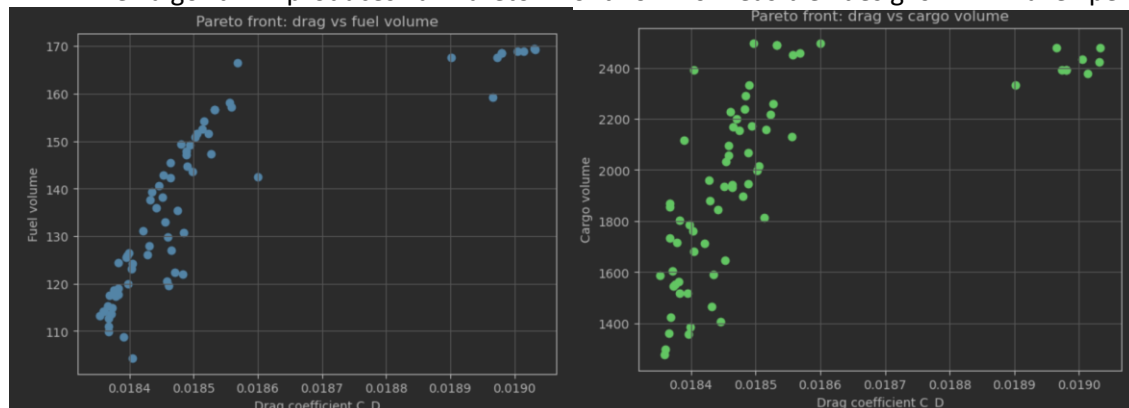
3. Parameter Choices

Population size: 70 — enough for diversity, fast enough to run in the notebook. Generations: 60 — the front stabilizes well before this. Crossover probability: 0.85 — high, since recombination is the main search mechanism. Mutation probability: 0.15 per variable, strength 0.07 of each variable's range — small enough not to destroy good solutions, large enough to avoid stagnation.

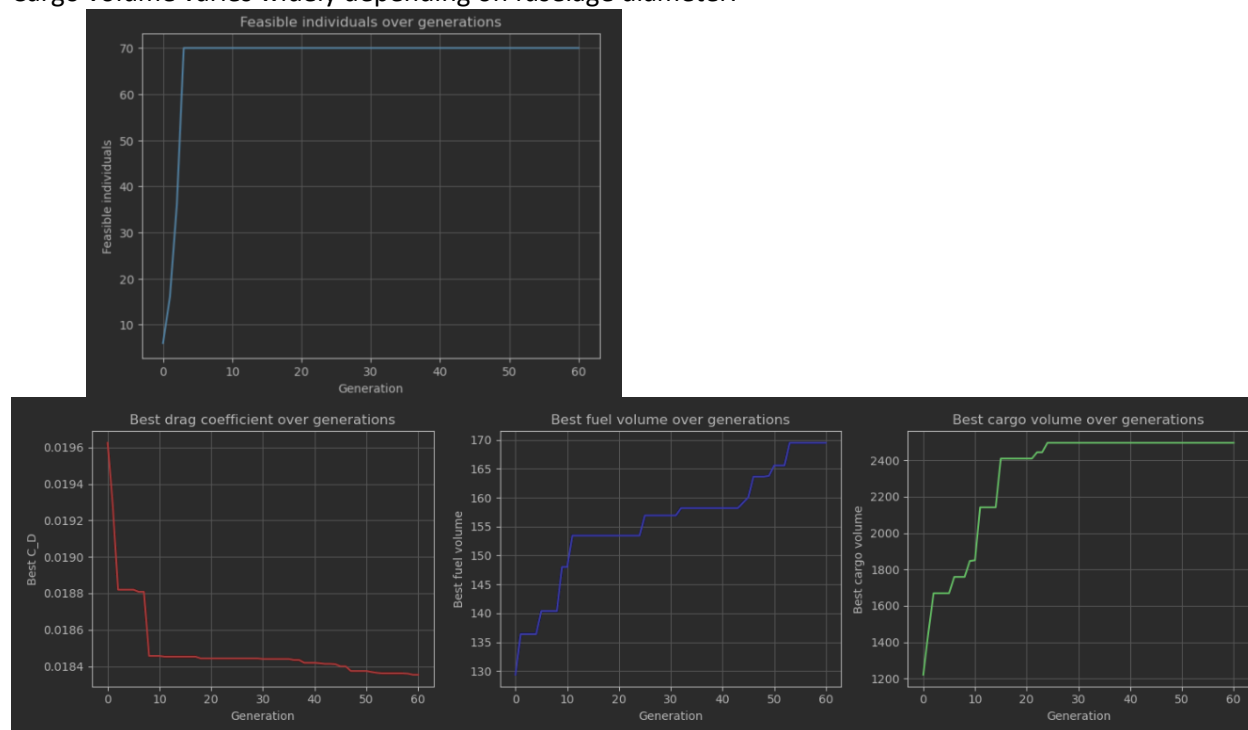
Variable bounds cover realistic aircraft geometries: $b \in [28, 70]$ m, $S_w \in [80, 300]$ m², sweep $\in [0^\circ, 35^\circ]$, $t_c \in [0.08, 0.18]$, $L_f \in [25, 65]$ m, $D_f \in [3, 8]$ m. No explicit penalty coefficient λ is used — squared violations are summed directly, and the feasibility rule handles the rest.

4. Results

The algorithm produces a Pareto front of 70 feasible designs. All have penalty = 0.



The plots show clear trade-offs. Designs with the highest fuel volume accept slightly more drag. Cargo volume varies widely depending on fuselage diameter.



The population becomes fully feasible within the first 10 generations. All three objectives improve steadily before stabilizing. The results table in the notebook shows 12 representative designs — wing area clusters near its upper bound and sweep near zero, which the model favors for low drag and high fuel volume.

5. Discussion

Wing area and fuselage diameter are the most influential variables. Wing area appears in both drag and fuel formulas. Fuselage diameter is squared in the cargo formula, so it has a disproportionate effect. Sweep has little influence in this simplified model.

The Pareto front is diverse in fuel (104–170 m³) and cargo (1277–2496 m³) but narrow in drag (0.0184–0.0190), which is a model artifact — once designs are feasible and at similar wing areas, drag doesn't vary much.

Limitations: the aerodynamic model is simplified, structural and stability constraints are not fully enforced, and all results depend on fixed physical constants. Possible improvements include adding an explicit lift check, increasing generations, or using a more realistic drag model.

6. Defense of Design Choices

A genetic algorithm with Pareto dominance is appropriate here because the problem is continuous, multi-objective, and constrained, and doesn't require gradient information. NSGA-II style selection with crowding distance is a standard choice for this type of problem.

The parameters were chosen to balance exploration and exploitation. Higher mutation would slow convergence or destroy good solutions; lower mutation would cause stagnation. The crossover probability of 0.85 reflects that recombination is the primary way to combine good designs. The main weakness of the approach is the simplified model — results may not transfer to a real design process.